

Sure Print Custom Shirt Website

Team 3.14159

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CS 191 WFQ1

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Customer Statement of Requirements

Project Motivation and Problem Description

Sure Print Custom Shirt is a small business that offers shirt printing and shirt-making services. They take in a large number of orders on a regular basis, and as their customer base continues to grow, so does the number of orders they need to handle on any given day. This growth is a good sign for the business, but it has also made their day-to-day operations harder to manage.

At the moment, the business does not use any kind of system or app to help them organize their orders. Customers reach out to them mainly through their Facebook page or by phone, and these same channels are also used to receive order details, confirm specifications, and follow up on order status. Once an order comes in, there is no centralized system in place to record and track it. Instead, the whole process is handled manually, often relying on disorganized notes, chat messages, or simply what the staff can remember.

Because of this manual setup, the business owners sometimes find themselves confused about which orders have already come in, what stage each order is currently at, and which ones still need attention. Without a clear and organized view of all their orders, it becomes easy to lose track of details, delay updates to customers, or accidentally overlook an order that is still in progress. As the number of orders increases, this problem only becomes harder to manage using memory and informal methods alone.

The goal of this project is to create a simple system that gives the business owners one central place to see, organize, and track all their orders, from the moment they are received to the moment they are completed. By addressing this core problem, the project aims to reduce confusion, prevent missed or delayed orders, and make it easier for the business to keep up with its growing number of customers.

Current Existing Solutions

In terms of just manual tracking and organizing orders, some people use Google Sheets or Microsoft Excel as a low-cost way to log and track basic information. While these tools can help to some extent, they are not built with the specific workflow of a print or shirt-making business in mind. They do not have a built-in way to represent production stages, such as design approval, printing, quality checking, and completion. They also do not have a structured way to capture order-specific details like size breakdown, materials, or design files, so this information usually ends up as plain text notes, which brings back the same risk of errors and miscommunication that the shop is currently experiencing.

There are already software solutions built specifically for print and apparel shops. Platforms like shopVOX, Printavo, Ordant, YoPrint offer features such as order tracking, quoting, invoicing, customer management, and even online portals where customers can place and proof their own orders. These tools show that the general problem of centralizing order and customer information for a printing shop is a known and common one, since many similar businesses run into the same manual tracking issues that this project aims to solve.

The dedicated printing shop platforms solve this workflow problem better, but they are usually built to serve a very wide range of businesses, from signage to embroidery to large commercial printing. Because of this, they often come with a lot of extra features that a small shop does not need, along with very expensive subscription costs, ranging around the \$100.00 – \$300.00 range (or ₱6000.00 – ₱18000.00 in peso). These platforms also have a learning curve that may not fit a business with non-technical staff and customers. This creates a gap between tools that are too generic to fit the workflow, and tools that are too complex or expensive for a small business's actual needs.

Key Requirements

- **Customer Portal**

- The customer should be able to place an order through the portal, specifying the type of service needed (printing or shirt-making).
- The customer should be able to input order specifications, such as design, size, quantity, and material.
- The customer should be able to upload a design file or reference image for their order, if applicable.
- The system should generate a unique order reference number automatically once an order is submitted, so the customer can use it to check their order later.

- **Order Tracking**

- The manager should be able to update the status of any order in the system (e.g., Order Received, In Production, Ready for Pickup, Completed).
- The customer should be able to view the current status of their order using their order reference number or contact details.
- The system should display orders in a way that allows the manager to quickly see which orders are pending, in progress, or completed.

- **Order History and Customer Database**

- The system should automatically store customer information (name, contact number, address) whenever a new order is placed.

- The manager should be able to look up a customer's past orders using their name or contact number.
- **Inventory Tracking**
 - The manager should be able to add, update, and remove inventory items (e.g., shirt materials, ink, sizes) in the system.
 - The system should automatically deduct inventory quantities when materials are used for a confirmed order.
 - The manager should be able to view current stock levels at any time.
- **Payment**
 - The manager should be able to record payment details for each order, including amount paid and payment method.

Preliminary UI Design

Note that the following designs are just mockups and are not the final designs / implementations for the project. These are just initial ideas of what the content of the pages should look like.

Customer Portal

SERVICE TYPE



Print on Shirt
You supply the shirt



Custom Shirt
We make and print

SIZE

XS

S

M

L

XL

XXL

3XL

QUANTITY

Select quantity

▼

PRINT SIDE

Front

Back

Both Sides

DESIGN FILE — optional



Drop file here or [browse](#)
PNG, JPG, PDF, AI, SVG — max 50MB

NOTES — optional

Special instructions, placement details, etc.

Place Order

Fill in service, size, quantity, and print side to continue

Order Tracking

Track Your Order

Enter your reference number or phone number

ORDER REFERENCE

SP-A3KX9M2P

— OF —

PHONE NUMBER

09XXXXXXXXX

Find Order

SP-A3KX9M2P

In Production

Service	Custom Shirt
Size	L
Quantity	20 pcs
Print Side	Front
Ordered	9/8/2026

Payment received — ₱3,200 via GCash

Order History

All 4

Order Received 1

In Production 1

Ready for Pickup 1

Completed 1

Search name, phone, ref...

REFERENCE	CUSTOMER	SERVICE	QTY	STATUS	PAYMENT	ACTIONS
SP-A3KX9M2P	Maria Santos 09171234567	Custom Shirt	20	In Production	₱3,200 (GCash)	Details
SP-BQ7YTR4L	Jose Reyes 09281239876	Print on Shirt	5	Order Received	Record Payment	Details
SP-C5NW6HJD	Ana Cruz 09339874321	Custom Shirt	50	Ready for Pickup	₱12,500 (Bank Transfer)	Details
SP-D2PF8KVN	Ramon Dela Cruz 09451237890	Print on Shirt	10	Completed	₱1,800 (Cash)	Details

Customer Database

4 customers on file

Search name or phone...

NAME	PHONE	ADDRESS	ORDERS	LAST ORDER	
Maria Santos	09171234567	123 Rizal St, Quezon City	1	9/8/2026	Hide
SP-A3KX9M2P	Custom Shirt	20 pcs · L	In Production	9/8/2026	
Jose Reyes	09281239876	45 Mabini Ave, Makati	1	9/10/2026	Orders
Ana Cruz	09339874321	78 Del Pilar, Pasig	1	9/6/2026	Orders
Ramon Dela Cruz	09451237890	12 Epifanio St, Manila	1	9/3/2026	Orders

Inventory

8 items in inventory

+ Add Item

MATERIAL

Item	Quantity	Unit	Low Stock At	Stock Level	
White Cotton Shirt (M)	120	pcs	20	OK	Edit Deduct Remove
White Cotton Shirt (L)	85	pcs	20	OK	Edit Deduct Remove
Black Dri-Fit Shirt (M)	8	pcs	15	Low Stock	Edit Deduct Remove

INK

Item	Quantity	Unit	Low Stock At	Stock Level	
DTG Ink — Black	4	liters	2	OK	Edit Deduct Remove
DTG Ink — Cyan	3.2	liters	2	OK	Edit Deduct Remove
DTG Ink — Magenta	1.1	liters	2	Low Stock	Edit Deduct Remove

Payment

TOTAL REVENUE

₱17,500

PAID ORDERS

3

PENDING PAYMENT

1

PAYMENT RECORDS

REFERENCE	CUSTOMER	AMOUNT	METHOD	RECORDED	ORDER STATUS
SP-A3KX9M2P	Maria Santos 09171234567	₱3,200	GCash	9/8/2026	In Production
SP-C5NW6HJD	Ana Cruz 09339874321	₱12,500	Bank Transfer	9/7/2026	Ready for Pickup
SP-D2PF8KVN	Ramon Dela Cruz 09451237890	₱1,800	Cash	9/6/2026	Completed

AWAITING PAYMENT

REFERENCE	CUSTOMER	SERVICE	QTY	STATUS
SP-BQ7YTR4L	Jose Reyes 09281239876	Print on Shirt	5	Order Received

II. Plan of Work

Estimates of Cost and Resources

The team consists of us three student developers working alongside academic coursework across the 12-week project lifecycle (September 13, 2026 – December 5, 2026). In line with academic project constraints, we will not be budgeting monetary compensation but instead will be tracked via person-hours.

Human Resources			
Member	Role	Responsibility	Est. Hours
Jason Patrick Fangonil	Full-Stack Developer/ Systems and QA Lead	Inventory tracking (auto-deductions), payment logging, integration testing, system deployment	6-10hrs/week
Ethan Kurt Espejo	Full-Stack Developer/ Frontend Lead	Customer portal (order intake, service selector, design upload), client-facing tracking UI	6-10hrs/week
Ambernie Salting	Full-Stack Developer/ Backend Lead	Relational database schema, API routes, reference code generation, order tracking engine	6-10hrs/week
Total Human Resources		~216 total hours	

System resources encompass the hardware, runtime environment, data storage, and software tools that we will need to develop, host, and operate the system. The project makes use of existing client hardware and developer free tiers to keep costs at minimum.

System Resources (Hardware & Software)			
Item	Technical Requirement	Purpose	Est. Cost
Admin / Shop Hardware	Dual-core CPU, 4 GB RAM, Windows 10+ / macOS / Linux	Running the manager dashboard to monitor orders, log payments, and track inventory. <i>(client has existing)</i>	₱ 0.0
Admin Display	Color display, minimum resolution 1280 × 720; web browser	Providing adequate screen estate to display order pipelines, inventory tables, and design previews. <i>(client has existing)</i>	₱ 0.0
Client / Mobile Devices	Android 8.0+ or iOS 12+ smartphone/tablet with modern mobile browser	Allows walk-in or remote customers to submit custom orders, upload logos, and check order status.	₱ 0.0
Cloud Hosting and Database	Cloud PaaS (Vercel / Render Free Tier)	Hosting the full-stack web application. Storing customer records, order line items, order status history, inventory balances, and payment transaction logs.	₱ 0.0
Network Bandwidth	Minimum 50mbps sustained internet connection at the shop location	Ensuring real-time synchronization of order status changes and reliable customer file uploads. <i>(client has existing)</i>	₱ 0.0
Payment Gateway	PayMongo Merchant Account	Processing digital downpayments and full settlements via QR Ph, GCash, Maya, and credit/debit cards.	₱ 0.0 (per-transaction fee only)
Custom Domain (Optional)	1-Year .com or .ph domain name registration (e.g., sureprintcustom.com)	Providing a professional, memorable URL for the customer order and tracking portal.	₱ 0.0-₱ 900.0
Total System Cost		₱ 0.0 - ₱ 900.0	

Schedules and Milestones

For this project, we will assume that there are 12 weeks, where Week 1 would be the week of September 13, 2026 – September 19, 2026, and Week 12 would be the week of November 29, 2026 – December 5, 2026.

Milestone	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Confirm details with the client												
Design database structure												
Build core database and backend												
Implement customer portal												
Implement order tracking												
Implement order and customer history												
Implement inventory tracking												
Implement payment module												
Software test report												
Handing over to client												

Project Metrics

To ensure the project is going well, the following metrics will be utilized:

Development and Process Metrics		
Metric	Description	Target Value
Schedule Variance	Actual Milestone Completion Date - Planned Milestone Date	<= 0 days
Requirements Volatility	Number of Changed or Added Requirements	< 15%

Development and Process Metrics		
Defect Density	Defects found in Testing over Function Points	< 0.15/FP

Operational, Usability, and Business Metrics		
Metric	Description	Target Value
Task Completion Time	Average time for staff to update an order stage or log a manual payment entry during user acceptance testing.	<= 20 seconds
Staff System Adoption Rate	Actual Person-Hours - Estimated Person-Hours	>= 90%
Payment Mode Fallback Utilization	Proportion of logged transactions handled via manual receipt upload vs. automated gateway checkout.	Monitored ratio
Data Re-Entry Count	Number of times a single order's details (sizes, cut, print placement, customer info) must be retyped from intake to completion.	1

Effort Breakdown

We divided the workload by assigning each module to whoever had the most experience or felt most confident handling it. For testing, we didn't assign a single owner because all of us will be working together on testing and bug fixing across the entire system.

Module	Components	Developer	Responsibility
Customer Portal & Assets	OrderSubmissionForm, ServiceTypeSelector, DesignUploadHandler, SpecificationMatrix	Ethan Kurt Espejo	Portal UI, interactive specification form, responsive canvas/file upload.
Core Backend & Database	OrderController, ReferenceCodeGenerator, CustomerRepository, DatabaseConnector	Ambernie Salting	Relational schema, database, reference code generator
Order Tracking & Status Pipeline	OrderStatusController, KanbanBoardView, OrderLookupService, TimelineLogger	Ambernie Salting (<i>Backend</i>) Ethan Kurt Espejo (<i>Frontend</i>)	Status transition state machine, manager order dashboard, customer lookup query service.
Customer & Order History	CustomerProfileController, OrderHistoryView, SearchFilterService	Ambernie Salting (<i>API/Queries</i>) Ethan Kurt Espejo (<i>UI/Search</i>)	Automatic customer ingestion upon submission, historical order lookup by phone/name, past transaction views.
Inventory Tracking & Auto-Deduction	InventoryController, MaterialItem, StockDeductionEngine, StockAlertBadge	Jason Patrick Fangonil	Inventory auto-deduction triggers on order confirmation, stock level views.
Payment Module	PaymentController, PaymentRecord, PaymentGatewayHandler, ReceiptProofUploader	Jason Patrick Fangonil	Manual payment logging, PayMongo integration

Project Risks

Take note of these legends for category and impact:

Category:

PS – Project size

BU – Business impact

TE – Technology to be built

DE – Development environment

ST – Staff size and experience

Impact:

1 – Catastrophic

2 – Critical

3 – Marginal

4 – Negligible

Risks	Category	Probability	Impact	RMMM
Requirements may change once the owners see the actual system	PS	70%	3	Do regular demos, log and schedule change requests for later phases.
Order volume may be higher than expected during peak seasons	PS	40%	3	Design for scalability, test with high-volume sample data.
Staff/owners may resist shifting to the new system	BU	30%	1	Involve staff in design, provide simple training after deployment.
Staff have little experience using computer-based systems	ST	70%	2	Keep UI simple, conduct training sessions.
Payment feature may not be fully implementable due to DTI/BIR requirements	BU	50%	2	Needs early communication with owners.
Design/UI may be too complex for non-technical users	DE	30%	2	Use simple UI patterns, and test with staff.
Data loss or corruption during development	TE	20%	1	Regular backups and version control, restore if needed.

References

shopVOX. (2026, May 5). Software for print, sign, Screen printing shops | Free trial. shopVOX.
<https://shopvox.com/>

Printavo. (2026). The easiest way to manage your print shop | Printavo.
<https://www.printavo.com>

Ordant. (2026). Print Estimating and Workflow Management Software | Ordant.
<https://ordant.com/>

YoPrint All-In-One Cloud Print Shop Management Solution. (2026). YoPrint.
<https://www.yoprint.com/>